LM-MPNN: An Integration Study of Language Models and Message Passing Networks on Quantum Chemistry

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Abstract

Pretrained language models (LMs) showcase significant capabilities in processing molecular text, while concurrently, message passing neural networks (MPNNs) demonstrate resilience and versatility in the domain of molecular science. Despite these advancements, we find there are limited studies investigating the relationship between molecular structures and their corresponding textual representations. Therefore, in this paper, we propose two strategies to evaluate whether an information integration can enhance the performance: contrast learning, which involves utilizing an MPNN to supervise the training of the LM, and fusion, which exploits information from both models. Our empirical analysis reveals that the integration approaches exhibit superior performance compared to baselines when applied to smaller molecular graphs, while these integration approaches do not yield performance enhancements on large scale graphs. Furthermore, we conduct experiments to assess the impact of dataset splitting strategies and random seeds on the overall performance.

1 Introduction

The success of attention mechanisms on tokens has introduced a massive family of language models based on Transformer architecture (Vaswani et al., 2017). Those large language models show their advantages in encoding sequential objects such as text (Liu et al., 2019), molecules (Honda et al., 2019), speech (Huang et al., 2021), and forecasting data (Giuliari et al., 2021). Particularly in the realm of molecules, pretrained molecule language models have exhibited some distinct capabilities in encoding chemical elements semantically without learning structures (Honda et al., 2019; Xia et al., 2022; Chithrananda et al., 2020; Wang et al., 2019a). Especially for proteins which function as natural components of the human body and a representative of molecule family, they could be efficiently encoded by transformer (Rao et al., 2019; Elnaggar et al., 2021; Rives et al., 2021; He et al., 2021) which acts as masked language modelers.

Unlike text, molecules possess an inherent relationship between elements, indicating the need for additional structural encoding besides word embeddings. The message passing neural network (MPNN), emerging as a prominent method for encoding structural information in recent years, demonstrates its robustness and versatility within the field of molecular sciences. By leveraging the 2-dimensional topological and 3-dimensional geometrical information as augmented features (Liu et al., 2021; Stärk et al., 2022), it is possible to learn molecular embeddings from structures without sequentially encoding traditional SMILES expressions.

The advent of MPNNs has promoted the exploration of graph-based learning methods for molecular science. Graph contrastive learning captures potential different structural distributions to fine-tune self-learned representations, where both local and global features are enhanced with chemical domain expertise (Stärk et al., 2022; You et al., 2021; Wang et al., 2022). Besides, the success of GPT (Radford et al., 2018) in traditional natural language processing tasks also motivates the research on graph transformers and graph GPT tailored for the molecule domain (Hu et al., 2020b; Bagal et al., 2021; Rong et al., 2020; Ying et al., 2021; Zhu et al., 2022).

In this paper, we aim to explore the interplay between molecular graph embeddings and SMILE token embeddings. We propose two categories of techniques for integrating information: contrast learning and fusion. In contrast learning-based methods, we incorporate an MPNN as an auxiliary model to supervise the training of the language

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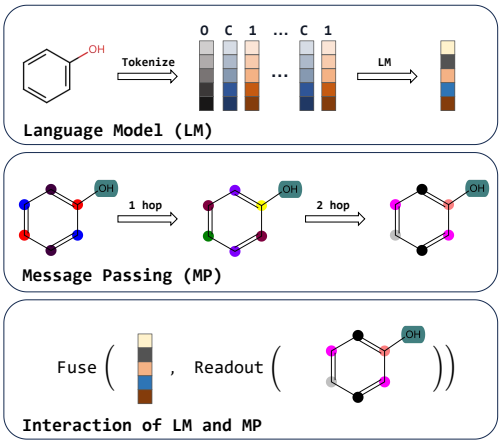

Figure 1: Baseline model: language model (LM) and message passing networks (MPNN). An interaction of LM and MPNN is investigated in this research.

model, operating at node or graph levels, while we only utilize the language model for downstream tasks. In fusion-based methods, we exploit information from both models to generate outputs for downstream tasks. This is achieved either by merging the output embeddings from both models or by integrating the output embeddings from one model with the input embeddings of the other.

Our main contributions are as follows:

1. Explore various information integration approaches to assess the necessity of incorporating supplementary structural features in molecular LLMs research, instead of pursuing a state-of-the-art performance.

2. Benchmark a series of combination of sequential-based methods (LM) and structural-based methods (MPNN) as baselines for further research.

2 Related Work

2.1 Molecule Representation Learning

The Simplified Molecular Input Line Entry System (SMILES) has become a cornerstone in cheminformatics, providing a compact and standardized representation for chemical structures. Conserving molecular structural information and atom orderings, the SMILES descriptor converts a molecule from its structural representation into a condensed 1-dimensional textual sequence. For example, a phenol molecule (C_6H_5OH) is represented as `C1=CC=C(C=C1)O`. Similar to the tokenization in natural language settings, a molecule is expressed as a sentence and atoms are expressed as words. This allows efficient utilization of large language models in chemical research.

2.2 Pretrained Large Language Models

The advent of the transformer architecture (Vaswani et al., 2023) represents a breakthrough in the field of natural language processing. Over the past few years, many excellent pretraining strategies have been proposed, such as BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019), significantly improving the capabilities of the large language models. As SMILES allows converting molecular structures into textual sentences, it is possible to apply language models for molecular machine learning, which facilitates the research on pretraining molecular language models.

Based on the implementation of RoBERTa, ChemBERTa (Chithrananda et al., 2020) employs masked-language modelling as its pretraining strategy, while the improved version ChemBERTa-2 (Ahmad et al., 2022) adopts multi-task regression as another pretraining task and uses larger training datasets. There are also other BERT-like transformer models, such as MolBERT (Fabian et al., 2020) and SMILES-BERT (Wang et al., 2019b), which are pretrained with different objectives on different datasets.

2.3 Contrastive Learning

Contrastive learning has emerged as a powerful paradigm in self-supervised learning. Unlike traditional methods that rely solely on labeled data, this approach leverages the differences between data to learn representations. Based on the assumption that similar instances should be closer in the embedding space, the objective is to maximize the similarity between positive data pairs while minimizing the similarity between negative data pairs. So far, contrastive learning has demonstrated efficacy across diverse domains. A common practice of this approach is based on data augmentation (You et al., 2021), where the utilization of unlabeled data enhances model generalizability and robustness. Furthermore, this approach is also widely adopted in the field of multimodality (Radford et al., 2021), where the availability of different data forms allows leveraging one representation to supervise the other.

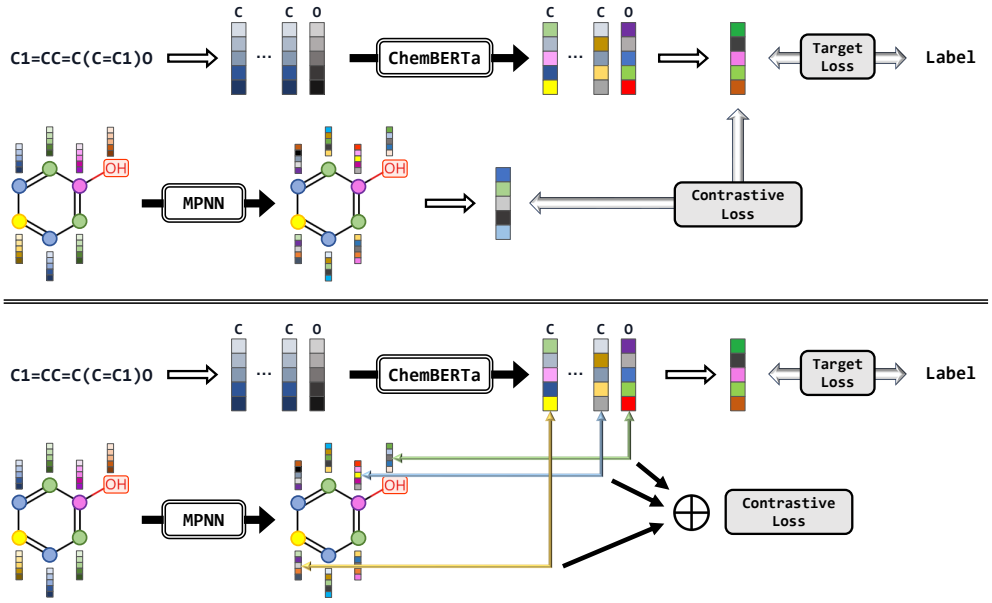

Figure 2: Contrastive Learning. Above: Node level contrastive learning. Below: Graph level contrastive learning.

3 Methods

We investigate two kinds of information merging methods: contrast learning-based methods and fusion-based methods. In contrast learning-based methods, we use GNN as an auxiliary model to supervise the training of the language model, while we only use the language model for downstream tasks. In fusion-based methods, we make use of information from both models to generate the output for downstream tasks. For each kind of method, we consider different model architectures. In this section, we will first briefly review the two baseline models ChemBERTa (Chithrananda et al., 2020) and NNConv (Gilmer et al., 2017), and then describe the contrast learning-based methods and the fusion-based methods.

3.1 Baseline (Figure 1)

Language Models. We choose ChemBERTa (Chithrananda et al., 2020) as our baseline model. The architecture of ChemBERTa is similar to BERT (Devlin et al., 2019), consisting of an embedding layer and several encoder layers. A molecule is first converted into the textual format through SMILES, and then a SMILES tokenizer is applied to convert the words into input tokens. After embedding lookup, each token is assigned with an embedding. Then the encoder layers which consist of a multi-head self-attention layer and a feed-forward layer transform the input token embeddings to hidden state representations. Finally, the task-specific output layer (classifier or regressor) predicts the result.

We let the ChemBERTa model produce three-level information for a molecule. First, node embeddings are extracted from the final hidden state representations. Since the SMILES transformation preserves atom orderings, each atom in the original molecule corresponds to a specific output token embedding. Second, the graph embedding is extracted from the special token at the beginning of the sequence. Third, the property prediction result is the final output.

The entire process can be depicted as follows:

$$
\begin{aligned}
&\texttt{Tokens} = \texttt{Tokenizer}(\texttt{Sequence}) \\
&E_{\text{in}} = \texttt{Embedding}(\texttt{Tokens}) \\
&E_{\text{out}} = \texttt{Encoder}(E_{\text{in}}) \\
&N = E_{\text{out}}[\texttt{Node_Indices}] \\
&G = E_{\text{out}}[0] \\
&P = \texttt{Predictor}(G)
\end{aligned}
\tag{1}
$$

where E_{in} and E_{out} represent the input token embeddings and final hidden state representations; N, G, and P represent the node embeddings, graph embedding, and property prediction result.

Message Passing Neural Networks. There are different types of graph neural networks, which include graph convolution, graph attention and neural message passing networks (MPNN). Edge at-

tributes or edge features are important in message passing mechanisms (Johannes et al., 2020; Gilmer et al., 2017). For the baseline model, we follow the model setting in the first paper of MPNN for Quantum Chemistry dataset QM9 (Gilmer et al., 2017), where it iteratively updated message m_v^t and hidden state h_v^t for each node v:

$$m_v^{(t+1)} = \sum_{u \in \mathcal{N}(v)} \texttt{Aggr}[h_v^{(t)}, h_u^{(t)}, e_{uv}] \quad (2)$$

$$h_v^{(t+1)} = \mathcal{U}(h_v^{(t)}, m_v^{(t+1)}) \quad (3)$$

$\texttt{Aggr}[\cdot]$ is the function that aggregated neighbor node u's information as well as the attributes e_{uv} of the shared edge with u. $\mathcal{U}(\cdot,\cdot)$ updates hidden states for v.

3.2 Integration 1: Contrastive Learning. (Figure 2)

Node Level Contrastive Learning It requires a triple ⟨**anchor, positive, negative**⟩ as an input to compute contrastive loss. Anchor and positive node embeddings are sampled from language models and message passing networks respectively. Negative samples are randomly generated from graphs with a different permutation. For one example node triple $t = \langle a, p, n \rangle =$ ⟨**anchor, positive, negative**⟩, where an example contrastive learning loss Triplet loss is given by:

$$L(t) = \max\{d(a,p) - d(a,n) + \text{margin}, 0\} \quad (4)$$

where margin is 1.0 and distance measurement $d(i,j)$ is defined as L_p-norm : $d(i,j) = \|i-j\|_p$. p is often set to 2 as an Euclidean distance metric. In a $\mathcal{M}$ mini-batch of training graphs (number of $\mathcal{N}$ nodes) with $\mathcal{K}$ triples, triplet loss is given by:

$$\text{L} = \sum_{m=1}^{\mathcal{M}} \sum_{i=0}^{\lceil \frac{\mathcal{N}}{\mathcal{K}} \rceil - 1} \sum_{j=1}^{\mathcal{K}} \max\{d(a_k^m, p_k^m) - d(a_k^m, n_k^m) + \text{margin}, 0\} \quad (5)$$

where $k = |\mathcal{K}|i + j$. Consider a binary classification problem, as mentioned we need to perform `Readout` function to obtain the global information of a graph. If its an average function, the total loss is given by a prediction loss for example negative log likelihood (NLL) loss, and a regularized triplet contrastive loss which is defined above:

$$\text{L} = \sum_{m=1}^{\mathcal{M}} \text{NLL}\left(\text{MLP}\left(\frac{1}{\mathcal{N}} \sum_{i=0}^{\lceil \frac{\mathcal{N}}{\mathcal{K}} \rceil - 1} \sum_{j=1}^{\mathcal{K}} a_k^m\right), y^m\right) + \alpha \cdot \sum_{m=1}^{\mathcal{M}} \sum_{i=0}^{\lceil \frac{\mathcal{N}}{\mathcal{K}} \rceil - 1} \sum_{j=1}^{\mathcal{K}} \max\{d(a_k^m, p_k^m) - d(a_k^m, n_k^m) + \text{margin}, 0\} \quad (6)$$

where $k = |\mathcal{K}|i + j$ likewise and α is a regularization term.

Graph Level Contrastive Learning. Apart from establishing negative samples between each pair of nodes at a fine grained level, we estimate contrastive learning at a coarse grained level which aims at computing difference between language model outputted graph embeddings and MPNN outputted graph embeddings. This could potentially avoid the situation that individual nodes with large difference contribute more to the difference of the molecule property. Moreover, the complexity is pretty lower than the complexity of node level comparison, which will be mentioned in part 3.4. Similar to the training loss function in (6), the graph level training loss is defined as:

$$\text{L} = \sum_{m=1}^{\mathcal{M}} \text{NLL}\left(\text{MLP}\left(a^m\right), y^m\right) + \alpha' \cdot \sum_{m=1}^{\mathcal{M}} \max\{d(a^m, p^m) - d(a^m, n^m) + \text{margin}, 0\} \quad (7)$$

where α' is a regularization term.

Note that different molecules could have a similar graph embedding which could lead to a similar quantum property, for example isomers. And also note that we use message passing network outputs to self-supervise (or fine-tune) language model outputs either in node level and graph level settings. It means that we didn't directly use MPNN outputs to perform predictions. This is because we want to see if injecting geometry information of molecules is beneficial to the end-to-end training of **language models**. It is different from the collaborative training (fusion) which would be introduced in below parts.

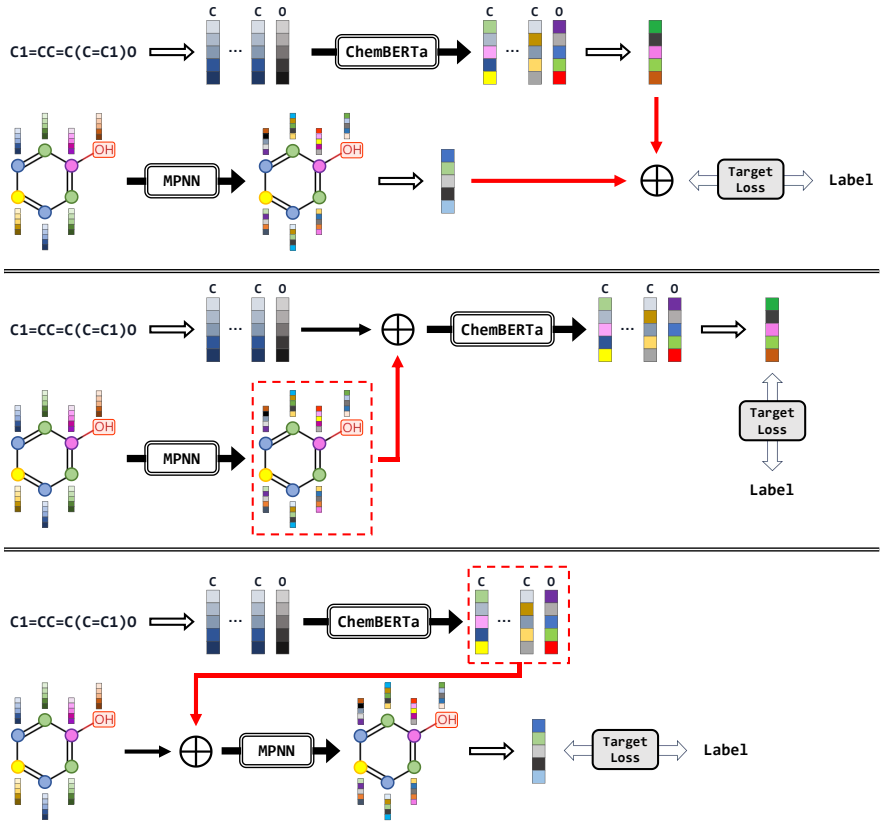

Figure 3: Fusion. Top: Late Fusion. Middle: MPNN2LM Joint Fusion. Bottom: LM2MPNN Joint Fusion.

3.3 Integration 2: Fusion (Figure 3)

Late Fusion. Different from self-supervised learning settings in part 3.2, we introduce another important interaction between LM embedding and MPNN embedding: late fusion (Sachan et al., 2021). It's called late fusion since their interaction happens after we extract their corresponding embeddings: h_{LM} and h_{MPNN}. The interaction is given by the notation $\bigoplus$. The prediction is then:

$$y_{\text{pred}} = \text{MLP}\left(h_{\text{LM}} \bigoplus h_{\text{MPNN}}\right) \tag{8}$$

$$\bigoplus = \{+, \max, \|, \odot\} \tag{9}$$

$$L = \sum_{i=1}^{\mathcal{M}} \text{NLL}(y^i_{\text{pred}}, y^i) \tag{10}$$

where the interaction contains element-wise addition, maximization, concatenation and gate function which is used for building highway layers.

Joint Fusion. Based on the late fusion, we consider two situations: 1) MPNN2LM: fuse initial graph embeddings and LM outputs $\Rightarrow$ perform MPNN downstream tasks, and 2) LM2MPNN: fuse initial token embeddings and MPNN outputs $\Rightarrow$ finetune LM downsteam tasks.

MPNN2LM. We initialized the word embedding for language model: $h^{(0)}_{\text{LM}}$. MPNN embedding is given by h_{MPNN}. Then the fused embedding is $h^{(0)}_{\text{LM}} \bigoplus h_{\text{MPNN}}$, which would be the input embedding for the pretrained language model. The node mask is also considered since in part 3.1 we mentioned that paddings are added to ensure the same length of input. After fine-tuning pretrained **LM**, we readout the global information h' to perform downstream tasks. Overall, the graph embedding prepared for MLP is:

$$h' = \texttt{Readout}\left(\textbf{LM}\left(h^{(0)}_{\text{LM}} \bigoplus h_{\text{MPNN}}, \textbf{mask}\right)\right) \tag{11}$$

LM2MPNN. Revisiting how MPNN works by (1) and (2), the node embedding for v are fused with the LM output for mask index v: $h^{(t)}_v \bigoplus h_{\text{LMM}(v)}$. Similar to the neighbor node u, they are fused with the LM output for mask index u: $h^{(t)}_u \bigoplus h_{\text{LMM}(u)}$.

Then the message is aggregated by:

$$m_v^{(t+1)} = \sum_{u \in \mathcal{N}(v)} \texttt{Aggr}\left(h_v^{(t)} \bigoplus h_{\text{LMM}(v)}, h_u^{(t)} \bigoplus h_{\text{LMM}(u)}, e_{uv}\right) \tag{12}$$

For the update function $\mathcal{U}(\cdot,\cdot)$, the new update rule:

$$h_v^{(t+1)} = \mathcal{U}\left(h_v^{(t)} \bigoplus h_{\text{LMM}(v)}, m_v^{(t+1)}\right) \tag{13}$$

3.4 Complexity Analysis

Baseline Models. The time complexity for baseline models are mainly dominated by their corresponding model architecture. Assume the input size $\in \mathbb{R}^{N \times d}$ For the pretrained transformer model, the self-attention module is the bottleneck, which is bounded by $O(N^2 \cdot d)$. Assume that there are L self-attention layers, then it will increase to $O(N^2 \cdot d \cdot L)$. The complexity for feed forward layers is $O(N \cdot d^2 \cdot L)$. The overall complexity for baseline LM is then $O(N^2 \cdot d \cdot L + N \cdot d^2 \cdot L)$. For MPNN, computing each message has a complexity of $O(d)$. The total complexity for the message passing step is then $O(E \cdot d)$. Updating nodes will be $O(N \cdot d^2)$. L layers lead to $O(L \cdot E \cdot d + L \cdot N \cdot d^2)$. So it depends on whether the graph is sparse or not. If graph is sparse, $N^2 \gg E$, then the complexity of LM is greater than the complexity of MPNN. This situation often occurs in real world applications.

Contrastive Learning. For contrastive learning, apart from the basic complexity for LM and MPNN, it also includes the complexity for computing contrastive loss. Take triplet loss as an example, in equation (5), the complexity is dominated by the last term. Assume that computing max only requires O(1). Computing $d(\cdot,\cdot)$ requires $O(d)$ if the dimension of inputs is d. Then the overall complexity is $O(d \cdot N)$ where N is the number of nodes in the graph. A node level contrastive learning then requires $O(N^2 \cdot d \cdot L + 2 \cdot N \cdot d^2 \cdot L + d \cdot N + L \cdot d \cdot E)$. For graph level contrastive learning we find that only complexity of model terms dominates, which leads to $O(N^2 \cdot d \cdot L + 2 \cdot N \cdot d^2 \cdot L + L \cdot d \cdot E + d)$, which is faster than that of node level.

Fusion We simply investigate $\bigoplus$ by choosing $\max$ which requires $O(1)$. The element wise maximization requires $O(N \cdot d)$ since input size is $N \times d$. Then the time complexity of late fusion would be the same as the complexity of nodel level contrastive learning, which is $O(N^2 \cdot d \cdot L + 2 \cdot N \cdot d^2 \cdot L + L \cdot d \cdot (N + E))$. `MPNN2LM` has the same complexity while `LM2MPNN` is much more complex since $\bigoplus$ directly affects the complexity of message passing operation. We already know that the complexity of MPNN is $O(E \cdot d)$. We assume that the average number of nodes is $\frac{2E}{N}$. Then the additional element wise addition contributions additional $O(E \cdot d)$, which leads to the overall complexity for `LM2MPNN`: $O(N^2 \cdot d \cdot L + 2 \cdot N \cdot d^2 \cdot L + L \cdot d \cdot (N + 2E))$.

4 Experiment settings

4.1 Dataset

We follow the following paradigm (Luo et al., 2022) for prediction on quantum chemistry based datasets: first we perform tests on small scale and classical benchmark molecule datasets, then we test its robustness on large scale and recently proposed benchmarks such as PCQM4Mv2 (Hu et al., 2020a). For small datasets we choose from MoleculeNet dataset (Wu et al., 2018) which collects data from physical chemistry, biophysics and physiology field. It has provided plenty of molecule datasets to play with (Wu et al., 2018). For large datasets, we choose **QM9** (Gilmer et al., 2017) as tested in MPNN. An extended (conference) version would be tested on a more large scaled dataset **PCQM4Mv2** (Hu et al., 2020a). The task is to predict property for each molecule using models in part 3. Selected datasets are **HIV**, **BACE**, **ESOL** and **BBBP** (Wu et al., 2018). A simple description of chosen dataset and task type is listed:

HIV, **BBBP**, and **BACE** are used for binary classification settings, while **ESOL** and **QM9** are used for regression settings. For simplicity, we only choose the first target from all 19 classes, which is the Dipole moment μ. For the regression problem, the performance is measured by mean absolute error (mae). As for the classification, it's measured by the mean accuracy (acc). Specifically, the pretrained ChemBERTa is time-consuming so instead we report a fixed accuracy with a fixed seed.

4.2 Hyper-parameter settings

There are two pretrained model to choose from: ChemBERTa and its improved version ChemBERTa-2. We choose Adam optimizer for optimizing model parameters with default learning rate 0.001 when running with pretrained ChemBERTa-2 (<4G). The initial learning rate is tuned to 0.0002 when running with ChemBERTa since the model size is large (>16G) which requires

Model	HIV (**acc.**) ↑	BACE (**acc.**) ↑	BBBP (**acc.**) ↑	ESOL (**mae.**) ↓
ChemBERTa baseline	97.55	**84.00**	**92.11**	0.545
MPNN baseline	97.94	67.00	81.58	0.550
ChemBERTa contra. MPNN (node)	97.90	**83.00**	90.13	**0.471**
ChemBERTa contra. MPNN (graph)	98.26	**81.00**	**92.11**	**0.460**
ChemBERTa + MPNN (graph)	98.02	**84.00**	90.79	0.475
ChemBERTa ← MPNN	97.90	80.00	**91.45**	**0.464**
ChemBERTa → MPNN	97.90	**83.00**	**94.08**	0.531

Table 1: Performance of pretrained ChemBERTa on MoleculeNet datasets.

Name	#graphs	#nodes	#features	#classes
HIV	41,127	∼25.5	9	1
BBBP	2,050	∼23.9	9	1
BACE	1,513	∼34.1	9	1
ESOL	1,128	∼13.3	9	1
QM9	130,831	∼18.0	11	19

a small learning rate. We follow a 9:1:1 train-valid-test ratio for MoleculeNet dataset, and follow an approximate 21:2:2 train-valid-test ratio for **QM9** dataset. Hidden dimension is set to 64. The default choice for $\bigoplus$ is sum (addition).

4.3 Scalability

A single NVIDIA A100 GPU could satisfy all our experiments. In other words, it is scalable for training all datasets including large scaled ones. It is also trainable on A100 for PCQM4Mv2 dataset according to the description of original OGB paper (Hu et al., 2020a). However, it is not scalable if we use SOTA graph transformer model in ablation study, so in ablation study we will simply use GraphGPS graph transformer (not yet done).

5 Results

Observation 0: Protein language models are more preferred. A basic observation is that message passing networks are generally inferior to language models in molecule property prediction. This phenomenon is also mentioned in the previous research work (Xu et al., 2022). This has indicated some works to include the geometric properties such as 3D information and rotation invariant parameters in message passing networks to reinforce its prediction and expressive power. The explanation of this phenomenon would be that 1) model size of ChemBERTa is much larger than message passing networks and 2) ChemBERTa has been pretrained on some more larger datasets for example ZINC dataset, while message passing networks do not follow the pretraining tradition of language models.

Observation 1: Integration on relatively small graphs are more preferred. Using the pretraind ChemBERTa-2, we found that both contrastive learning and fusion methods outperform baseline models in **ESOL**, **BACE**, and **BBBP** where they are relatively small compared with **QM9** and **HIV** datasets. Especially, graph level contrastive learning performs the best and it seems to be robust among all tasks, followed by node level contrastive learning and late fusion methods. In large dataset, the tuning strategy might influence the potential performance, where it splits the dataset in a better way therefore we perform one ablation regarding train test split (in section 6) to avoid the difference that brought by dataset itself.

Observation 2: Integration w.r.t both regression and classification are useful. In terms of training convergence, we observe that the accuracy or mean absolute error converges quickly to a high or low score respectively. For small graph datasets **BACE** and **BBBP** on graph classification problem, an improvement of **0.032** on average accuracy is observed with method `MPNN2LM`. For small graph dataset **ESOL** on regression problem, an improvement of **0.035** is observed with method graph level contrastive learning. For **HIV**, we observe a little improvement with node level contrastive learning. For **QM9**, there is almost no improvement observed. This is because for QM9 and pretrained model ChemBERTa, we only run once since it takes almost a day to finish running one large dataset with a single A100 without any hyper-parameter tunning. Also, we do not consider the SOTA MPNN and pretrained molecule language models, therefore there is a small gap in performance table between the proposed methods and baseline. We will fix this in the extended version.

Observation 3: Pretrained language models are important for downstream predictions. In comparison to ChemBERTa-2, our proposed model with ChemBERTa pretrained performs worse with

Model	HIV (**acc.**) ↑	BACE (**acc.**) ↑	BBBP (**acc.**) ↑	ESOL (**mae.**) ↓
ChemBERTa-2 baseline	98.06 ± 0.002	84.60 ± 0.014	91.97 ± 0.025	0.463 ± 0.016
MPNN baseline	97.94	67.00	81.58	0.550
ChemBERTa-2 contra. MPNN (node)	98.07 ± 0.002	$\mathbf{86.80 \pm 0.024}$	$\mathbf{94.34 \pm 0.017}$	0.447 ± 0.020
ChemBERTa-2 contra. MPNN (graph)	97.98 ± 0.002	$\mathbf{86.80 \pm 0.026}$	$\mathbf{94.34 \pm 0.012}$	$\mathbf{0.428 \pm 0.040}$
ChemBERTa-2 + MPNN (graph)	98.04 ± 0.002	$\mathbf{86.80 \pm 0.024}$	$\mathbf{93.82 \pm 0.012}$	$\mathbf{0.431 \pm 0.026}$
ChemBERTa-2 ← MPNN	98.00 ± 0.001	$\mathbf{87.80 \pm 0.026}$	91.31 ± 0.009	0.459 ± 0.031
ChemBERTa-2 → MPNN	97.95 ± 0.003	85.80 ± 0.021	93.16 ± 0.008	$\mathbf{0.444 \pm 0.021}$

Table 2: Performance of improved pretrained ChemBERTa-2 on MoleculeNet datasets.

a fixed seed. Our suggestion is that researchers should first pick a list of pretrained models and integrate them in order to choose the most appropriate pretrained language model for further training, followed by an investigation (prompting) with foundation models such as GPT4.

Observation 4: Joint Fusion to some extent helps learn individual model better. We also focus on if such multi-modal module (Figure 2, Figure 3) helps learn individual module (Figure 1) better. It improves a lot for single MPNN baseline if we consider its language level information as augmented features. For example, for **BACE** dataset, MPNN has an accuracy of 0.67. With injecting pretrained language information, an improvement of 0.188 is observed. Single pretrained ChemBERTa-2 has an acuuracy of 0.846. With injecting structural information, an improvement of 0.032 is observed. In small datasets aiming at regression, this phenomenon also occurs.

Model	QM9 (target = 0)
ChemBERTa-2 baseline	**0.4601**
MPNN baseline	**0.4607**
ChemBERTa-2 contra. MPNN (node)	0.4989
ChemBERTa-2 contra. MPNN (graph)	0.4721
ChemBERTa-2 + MPNN (graph)	0.4776
ChemBERTa-2 ← MPNN	0.5231
ChemBERTa-2 → MPNN	**0.4716**

Table 3: Performance of improved pretrained ChemBERTa-2 on QM9 dataset.

6 Ablation Study

Effects of datasets. We choose another dataset in MoleculeNet to certify that the proposed models are still robust on this dataset. Take **FreeSolv** as an example, we figure out that graph level contrastive learning is still robust on this regression task. `LM2MPNN` performs the best, which improves by approximately 0.04 from the baseline language model ChemBERTa-2. Surprisingly we found that `MPNN2LM` doesnt work well. This similar pattern could also be observed from table 2 and table 3, where we get into a conclusion that injecting structural information into word embeddings would be a bad idea and we suggest that researchers should be careful when fine-tuning the individual language model with additional structural features.

Model	FreeSolv (**mae.**) ↓
ChemBERTa-2 baseline	0.5058
MPNN baseline	0.9305
ChemBERTa-2 contra. MPNN (node)	0.5035
ChemBERTa-2 contra. MPNN (graph)	**0.4786**
ChemBERTa-2 + MPNN (graph)	**0.4758**
ChemBERTa-2 ← MPNN	0.5789
ChemBERTa-2 → MPNN	**0.4646**

Table 4: Performance of improved pretrained ChemBERTa-2 on FreeSolv dataset with fixed seed 42.

Effects of dataset split. We want to figure out if split of training, validation and test datasets lead to different performance. We run on **BBBP** (classification) and **ESOL** (regression). Four ratios are considered: 9:0.5:0.5, 8:1:1, 7:2:1, and 6:2:2. Model prediction power is highest at a ratio of 8:1:1.

Train test split	BBBP (**acc.**) ↑	ESOL (**mae.**) ↓
9: 0.5 : 0.5	94.74	0.5793
8 : 1 : 1	**94.74**	**0.3585**
7 : 2 : 1	94.08	0.4902
6 : 2 : 2	93.43	0.4991

Table 5: Model (graph level contrast.)

Effects of different fusion operation $\bigoplus$ We first follow the default train valid test split of 8:1:1. As mentioned, there are four fusion operations $\bigoplus$, which are max, sum, concatenation and gate function. Concatenation and gate function are better fusion choice for **BBBP** while addition and maxmization are better fusion choice for **ESOL**. These fusion preference could be extended to corresponding classification and regression tasks.

Effects of different graph neural networks As mentioned in section 3, there are three types of graph neural networks in mainstream GNN research, which are graph convolution (GraphConv),

Fusion Operation	BBBP (**acc.**) ↑	ESOL (**mae.**) ↓
sum	94.74	**0.3585**
max	94.74	**0.3569**
concate	**96.05**	0.3851
gate	**95.39**	0.3702

Table 6: Model: Late Fusion

message passing neural networks (MPNN), and graph attention networks. We substitute MPNN with with GraphConv to see if MPNN is much better than other types of GNN for baselines. The results show that MPNN is more preferred to **BBBP** and **ESOL** since it includes edge attributes as additional information.

Fusion Operation	BBBP (**acc.**) ↑	ESOL (**mae.**) ↓
MPNN	**94.74**	**0.4317**
GraphConv	**92.11**	**0.5195**

Table 7: Model: Late Fusion (`MPNN2LM`)

7 Conclusion

In this paper, we delved into various information integration approaches to assess whether the collaborative utilization of language models (LMs) and message passing neural networks (MPNNs) surpasses the individual efficacy of these models. We evaluated the integration approaches over different graph scales on both classification and regression tasks. Our empirical analysis has demonstrated that the integration approaches outperform the baselines on small-scale graphs but do not yield improvements on datasets of larger scales. Furthermore, we have found that differences in dataset splitting strategies, random seeds, and aggregation choices in fusion all have an impact on the overall performance.

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